A Novel Leakage Model in OpenSSL's Miller-Rabin Primality Test

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Abstract. At CRYPTO 2009, Heninger and Shacham presented a branch-and-prune algorithm for reconstructing an RSA private key given a random fraction of its private components. This method has been widely adopted in side-channel attacks, and its complexity is closely related to the specific leakage pattern encountered. In this work, we propose a new leakage model that enables efficient key reconstruction by exploiting a vulnerability in the modular exponentiation invoked by OpenSSL's implementation of Miller-Rabin primality test. This vulnerability reveals the least significant b bits of each window. Through our proposed model, these bits form new leakage patterns, which we term aligned and misaligned. In particular, the misaligned case includes previously undocumented scenarios where full key recovery is achievable without branching. Then we analyze the global and local behavior of key reconstruction under these patterns. Our evaluation demonstrates that they yield more efficient key reconstruction and maintain this advantage even in the presence of additional erasures. Moreover, in specific scenarios, successful reconstruction remains practical even if the bits obtained are less than 50%. Finally, we conducted a series of experiments to confirm the practicality of our work, successfully recovering the lower 4 bits from each 6-bit window and demonstrating efficient key reconstruction under our model.

Keywords: Partial key exposure attacks · Primality test · Modular exponentiation · Power analysis.

1 Introduction

RSA is a mainstream public key cryptosystem that has provided various cryptographic services for decades. Its security relies on the practical difficulty of integer factorization and has been intensively discussed by the community. One research direction pursued by cryptanalysis is to examine this assumption when side information is available, known as partial key exposure attacks, which seek efficient key reconstruction given a fraction of the secrets. To carry out real-world attacks, a growing field of research has emerged known as side-channel attacks (SCAs), drawing inspiration from Kocher's seminal works [31, 32]. This area leverages physical information from running cryptosystems, including power, electromagnetics, time, sound, cache footprints, etc., to extract sufficient secret bits. Some of these attacks have considerable strength, allowing them to derive the full key based solely on the physical information obtained [29, 35, 36, 40]. In practice, noise and countermeasures usually result in incomplete key acquisition, and partial key exposure attacks are required to finish the analysis [7, 8, 50, 51].

1.1 Partial Key Exposure Attacks From Non-Consecutive Bits

Ideally, cryptosystems are expected to offer a certain security strength even if partial key bits are compromised, but it is well known that RSA does not enjoy this property. For security purposes, it is critical to know how many partial bits of the secret components suffice to break the system.

Let the public key be (N, e) and the private key be (p, q, d), with $N = pq$ and $d \equiv e^{-1} \mod (p-1)(q-1)$. n denotes the length of p and q. In 1985, Rivest and Shamir [44] indicated that given 2/3 least significant bits (LSBs) of p or q is sufficient to break RSA in polynomial-time. In Coppersmith's pioneered work [19], this result was improved to 1/2 of the most significant bits (MSBs) or LSBs of a factor by utilizing the lattice reduction technique. Later, Howgrave-Graham [27] presented an alternative method to settle this problem by introducing the dual lattice. In 1998, Boneh et al. [10] presented a polynomial-time attack when a quarter of the LSBs of d are given. As the Chinese Remainder Theorem (CRT) is often used to speed up RSA implementation, in 2003, Blömer and May [9] investigated partial key exposure attacks on CRT-RSA and stated that known half of the bits of $d_p = d \mod (p-1)$ or $d_q = d \mod (q-1)$ could efficiently factorize N.

In 2008, Herrmann and May [26] demonstrated that the problem of factorizing N could be solved in polynomial-time when given at most $\mathcal{O}(\log\log N)$ blocks of bits. Concurrently, Halderman et al. [23] proposed cold boot attacks and introduced a new leakage pattern that reveals a random fraction of secret bits. According to the bound presented in [26], lattice-based techniques are not directly applicable to this case. In 2009, Heninger and Shacham [25] explored substantial redundancy in RSA key storage. Based on cold boot attacks, they developed a branch-and-prune algorithm that could efficiently recover the private key given a random fraction of p, q, d, d_p, d_q.

Heninger and Shacham considered an idealized scenario in which all given information is error-free, later termed the erasure model. Under this assumption, their algorithm reconstructs the key deterministically. However, when the given bits contain errors, this algorithm fails as the correct solution is inevitably pruned during the process. In 2010, Henecka et al. [24] proposed a probabilistic algorithm capable of correcting a secret key corrupted by random bit-flip errors. This work was subsequently refined by Paterson et al. [42] with their asymmetric bit-flip model. Later, Kunihiro et al. [33] analyzed the key reconstruction in the presence of both errors and erasures. Saito et al. [45] adapted Henecka's algorithm to efficiently handle leakage in windowed exponentiation.

1.2 Side-Channel Attacks on RSA Modular Exponentiation

Numerous attacks have been proposed to exploit vulnerabilities in RSA implementations, including but not limited to modular exponentiation [1, 3–5, 8, 11, 14, 15, 21, 22, 28, 29, 34–37, 39, 43, 45–47, 50–52], binary GCD computation [7], modular inversion [2, 6, 40], key validation [49] and pseudorandom number generators [17, 18].

Modular exponentiation is a core operation in the entire life cycle of RSA, spanning key generation, encryption, and decryption. Its implementation in cryptographic libraries has undergone rigorous scrutiny and several significant iterations to enhance security and reliability. The square-and-multiply method, which exploits the binary representation of the exponent, is less efficient and has been proven to be insecure in Kocher's timing attack [31]. To improve performance, modern open-source cryptographic libraries employ window exponentiation techniques, including fixed-window exponentiation (as in OpenSSL) and sliding-window exponentiation (as in Libgcrypt).

However, windowed exponentiation alone cannot guarantee constant-time execution, as loading precomputed multipliers remains vulnerable to cache-based attacks [8, 43, 51]. To mitigate this issue, a dummy load technique has become a widely adopted countermeasure. This method works by loading all precomputed operands and then selecting the desired value through bitwise masking operations, thereby eliminating observable timing differences. In [5, 28, 45], single-trace attacks are presented against this countermeasure. Specifically, Alam et al. [5] circumvented this countermeasure by targeting the exponent extraction in OpenSSL's modular exponentiation. Saito et al. [45] focused on the GNU Multiple Precision Library, developing a deep learning based classifier that uses power analysis to reliably distinguish secret-dependent memory accesses from dummy operations. Hu et al. [28] demonstrated that the dummy load technique introduces measurable power differences during mask computation, allowing efficient key extraction in the sliding-window exponentiation implemented within Libgcrypt.

1.3 Motivation

Lattice-based RSA key reconstruction methods typically require consecutive known bits of the private key [9, 10, 19, 26, 27, 44]. In comparison, the branch-and-prune algorithm introduced by Heninger and Shacham effectively handles randomly distributed known bits, rendering it suitable for SCAs [25]. While originally proposed in the context of cold-boot attacks, this algorithm has been applied to a variety of side-channel leakage settings [8, 16, 40, 51]. Heninger and Shacham [25] demonstrate that under a random leakage model, key reconstruction is achievable in polynomial time once a certain fraction of private key bits is available. The complexity of this process decreases as more bits are revealed. However, it should be noted that the efficiency of the branch-and-prune algorithm depends not only on the number of known bits but also by their distribution, that is, the specific leakage pattern.

Given the same amount of known bits, the complexity of key reconstruction via the branch-and-prune algorithm differs significantly across different leakage patterns. Consider an adversary who obtains 50% of the bits of p and q. If these bits form a block of consecutive least significant bits, the branch-and-prune algorithm faces significant computational challenges. Under the constraint $N = pq$, knowledge of either $p[i]$ (the i-th bit of p) or $q[i]$ is sufficient to uniquely determine the other. In this scenario, the redundant information concentrated in the lower bits offers nothing for determining the higher bits during the branch-and-prune process. As a result, the algorithm need to examine up to $2^{n/2}$ candidates to recover the remaining bits, making the reconstruction process computationally infeasible in practice. However, if the known bits follow a distribution, where at each index i, exactly one of $p[i]$ and $q[i]$ is known, the branch-and-prune algorithm can reconstruct the key without generating false candidates. Consequently, its execution becomes highly efficient, with negligible computational cost.

The behavior of Heninger and Shacham's algorithm described above motivates further investigation of new leakage in real-world implementations. OpenSSL, a widely deployed cryptographic library, prevents timing leakage by employing multiple mechanisms to achieve constant-time modular exponentiation, including: (1) a fixed-window method to prevent leakage through square-and-multiply chain; (2) padding the most significant bits of the exponent with zeros to avoid leaking its higher bits; (3) a constant-time look-up table using dummy load operations. The combination of these countermeasures increases the difficulty of attacking modular exponentiation in OpenSSL. Nevertheless, OpenSSL's implementation of modular exponentiation remains vulnerable to other forms of leakage, such as power consumption. When examining the modular exponentiation invoked during primality testing in key generation, an interesting form of leakage emerges. Through our proposed leakage model, we identified new leakage scenarios, including cases where the branch-and-prune algorithm recovers the key without generating any false candidates, which is observed for the first time in a real-world setting.

1.4 Our Contributions

The main contributions of this work are summarized as follows:

- We propose a novel leakage model that enables RSA key reconstruction with the branch-and-prune algorithm by exploiting a previously unnoticed leakage in OpenSSL's Miller-Rabin primality test. First, we reveal a vulnerability in OpenSSL's constant-time modular exponentiation that allows an adversary to extract b bits from each w-bit window during execution. When this operation is invoked by the Miller-Rabin test, it leaks partial information about p and q. Our model maps these leaked bits to an exploitable form. The partial information of p and q is then partitioned into three blocks, among which the leakage pattern of **Block 3** governs the complexity of the key reconstruction process. It comprises two patterns: aligned and misaligned, with the latter including a notable scenario in which the key can be rebuilt without generating any false candidates.

– We characterize the complexity of key reconstruction in aligned and misaligned cases by evaluating the number of candidate solutions examined. To this end, the branching behavior of the branch-and-prune algorithm within a single window (local) and across the entire process (global) is analyzed. We derive the mean and variance of the total number of candidates examined during the reconstruction process and establish an estimated upper bound. Further discussion is instantiated with specific OpenSSL-based parameters: $b = 4$, $w = 6$ for RSA-2048/4096; and $b = 2$, $w = 4$ for RSA-1024. The efficiency observed in misaligned cases, particularly when reconstructing the key without branching, suggests potential tolerance for additional erasures. Our analysis confirms that the speed advantage in misaligned cases persists even with the introduction of additional erasures.
– We conducted practical simple power analysis (SPA) attacks targeting the fixed-window modular exponentiation in OpenSSL's Miller-Rabin primality test, running on an ARM Cortex-M4 processor. The targeted implementation uses dummy load operations to access a pre-computed table in constant time. For a 6-bit window, the first two bits determine the row of the target element in this table, and the remaining four bits determine its column. Table lookup performs 16 iterations. In 15 of these, an all-0 mask is applied, and only the iteration corresponding to the target element uses an all-1 mask. This creates a detectable difference in power consumption between the target iteration and the others. By exploiting this leakage, we successfully recovered the least significant 4 bits from each 6-bit window during a single execution of the modular exponentiation. These recovered bits were then used in our proposed model to perform RSA key reconstruction. Furthermore, we analyzed the occurrence likelihood of the proposed scenarios by repeatedly invoking OpenSSL's key generation, confirming that misaligned scenarios exhibit non-negligible occurrence probabilities.

OpenSSL Long-Term Support Version. Notably, OpenSSL currently maintains two long-term support (LTS) versions: OpenSSL 3.0 and OpenSSL 3.5, both of which generate probable primes with conditions based on auxiliary probable primes [41]. The only difference lies in the number of Miller–Rabin tests (or modular exponentiation calls) performed. Taking RSA-2048 as an example, OpenSSL 3.0 [48] executes 64 tests on each auxiliary probable prime, as well as on the resulting primes p and q. In contrast, OpenSSL 3.5, updated from NIST FIPS 186-5 standard [41], performs 38 tests on auxiliary primes and only 5 tests on p and q. Nevertheless, these differences do not affect the applicability of our method, since our SPA attack relies solely on a single execution of the modular exponentiation to extract the bits required for key reconstruction. For demonstration purposes, all OpenSSL source code presented in the following sections is taken from the latest version of the 3.0 series.

Conclusion. The vulnerability in modular exponentiation exploited in this work is also applicable to attacks on RSA decryption. In practice, the widespread use of the Chinese Remainder Theorem in RSA implementations indicates that such attacks often result in partial leakage of d_p and d_q. As demonstrated in CacheBleed [51], reconstructing the private key based on partial bits of d_p and d_q requires guessing both k_p and k_q, where $ed_p = 1+k_p(p-1)$ and $ed_q = 1+k_q(q-1)$. In the common case where $e = 65537$, this involves searching through 65,537 possible pairs of k_p and k_q [51], significantly increasing attack complexity.

In conclusion, attacking the modular exponentiation in the Miller-Rabin primality test within OpenSSL's RSA implementation (as opposed to other stages) provides the following advantages: (1) Key reconstruction benefits from our new leakage model, with improved efficiency and additional erasure tolerance. (2) Adversaries recover p and q directly, bypassing the complexities associated with implementations of the Chinese Remainder Theorem.

1.5 Paper Organization

The remainder of the paper is organized as follows. Section 2 gives the preliminaries of the leakage model and describes Heninger and Shacham's work. Section 3 details the vulnerability in the Miller-Rabin primality test implemented in OpenSSL and introduces our proposed leakage model. In Section 4, we conducted a comprehensive evaluation of the global and local behavior of the key reconstruction based on the new model, including analysis of scenarios with additional erasures. Section 5 demonstrates our practical attack and experimental results. Section 6 offers conclusion and implications.

2 Background

In this paper, we denote variables by uppercase italic letters (e.g., X) or lowercase italic letters (e.g., x). Upright uppercase letters, such as $\mathrm{G}(\cdot)$ denote the probability generating function (PGF), while $\Pr(\cdot)$, $\mathrm{E}(\cdot)$, and $\mathrm{Var}(\cdot)$ represent probability mass function, expectation, and variance, respectively. $p[i]$ represents the i-th bit of variable p, and $p[i:j]$ denotes the bits from the j-th bit (lower) to the i-th bit (higher) of p, inclusive of both $p[i]$ and $p[j]$.

2.1 Fixed-Window Modular Exponentiation in OpenSSL

Several algorithms have been suggested for RSA modular exponentiation, including the square-and-multiply, the sliding-window, and the fixed-window approach. The latter has gained widespread adoption in cryptographic libraries due to its inherent feature against leaking the square and multiplication sequence.

Given a fixed window of size ω, a base a, a modular N, and an n-bit exponent p denoted as a sequence of digits in base 2^ω, fixed-window exponentiation outputs $r = a^p \mod N$ using two steps. It first computes a set of multipliers $a_j = a^j \mod N$ for $0 \leq j \leq 2^\omega - 1$. Then from the most significant to the least significant digit of p, r is accumulated by squaring ω times and multiplying a_{p_i} in each iteration.

A naive implementation of fixed-window exponentiation suffers from severe lookup table leaks, especially from cache-based channels. OpenSSL mitigated this vulnerability by integrating dummy load operations with a scatter-gather memory layout, following Intel's recommendation [12, 13]. This technique partitions each precomputed multiplier into blocks and interleaves blocks from different multipliers within the same cache line, thereby making cache-line access patterns independent of the specific multiplier being processed. Under this memory layout, identifying specific cache line accesses could leak partial bits of the window exponent, necessitating dummy loads to obscure cache access patterns. Building on this implementation, Yarom et al. [51] devised CacheBleed, a sub-cache-line attack capable of exposing the least significant three bits of each window through cache bank conflicts. In response, OpenSSL 1.0.2g implemented a patch that iterates over all chunks of each multiplier and applies a masking operation for selection. This countermeasure remains active by default in current OpenSSL releases.

2.2 Miller-Rabin Primality Test

Cryptographic libraries typically call a random number generator followed by a primality test to obtain secret components p and q. Miller-Rabin test, a probabilistic algorithm that returns whether a given number is likely to be prime, is frequently employed to check for prime numbers.

Given a prime $p = 2^s \cdot p' + 1$ and an integer base a such that $0 < a < p$, then one of the following congruence relations holds: $a^{p'} \equiv 1 \mod p$ and $a^{2^r \cdot p'} \equiv -1 \mod p$ for some $0 \leq r < s$. This can be proved based on two facts:

- Fermat's little theorem: $a^{p-1} \equiv 1 \mod p$;
- The only square roots of 1 modulo p are 1 and -1.

However, the inverse of the above property is incorrect. One of the above congruence relations holding does not guarantee p is prime. It only promises such p is a strong probable prime to base a. A small fraction of composites, known as strong pseudoprimes, also pass this test. Miller-Rabin algorithm assesses the primality of a given number with different bases. If enough tests are passed, the number is said to be prime with high probability. The number of tests is determined according to the size of p and the desired probability.

2.3 Probability Generating Functions [30]

The probability generating function refers to the power series expression of the probability mass function of random variables. It is often used to describe the probability distribution of a discrete random variable.

Definition 1 (Probability Generating Functions). *Let X be a discrete random variable taking values in the non-negative integers. The PGF of X is defined as*

$$\mathrm{G}_X(s) = \mathrm{E}(s^X) = \sum_{x=0}^{\infty} s^x \Pr(X = x).$$

Algorithm 1 Miller-Rabin Primality Test

Require: prime candidate $p = 2^s \cdot p' + 1$, number of rounds j,
Ensure: statement "p is composite" or "p is likely prime".

for $i = 1$ to j **do**
  choose a random $a \in \{2, 3, ..., p-2\}$;
  $u = a^{p'} \mod p$;
  if $u \neq 1$ and $u \neq p-1$ **then**
    for $j = 1$ to $s-1$ **do**
      $u = u^2 \mod p$;
      if $u = 1$ **then**
        return ("p is composite")
      end if
    end for
    if $u \neq p-1$ **then**
      return ("p is composite")
    end if
  end if
end for
return ("p is likely prime")

The following properties can be obtained according to the definition of PGF, expectation, and variance: (1) $\mathrm{G}_X(1) = \sum_{x=0}^{\infty} \Pr(X = x) = 1$, (2) $\mathrm{E}(X) = \mathrm{G}'_X(1)$ and (3) $\mathrm{Var}(X) = \mathrm{G}''_X(1) + \mathrm{G}'_X(1) - \mathrm{G}'_X(1)^2$.

Theorem 1. *Assume that X_1,...,X_n are independent random variables, and $Y = X_1 + ... + X_n$. Then*

$$\mathrm{G}_Y(s) = \prod_{i=1}^{n} \mathrm{G}_{X_i}(s).$$

Theorem 2. *Assume that $\{X_i\}$ is a sequence of independent and identically distributed random variables with the common PGF G_X. Let $Y = X_1 + ... + X_M$, where M is a random variable independent of X_i. Denoting the PGF of M as G_M, the PGF of Y is*

$$\mathrm{G}_Y(s) = \mathrm{G}_M(\mathrm{G}_X(s)).$$

2.4 The Heninger-Shacham Algorithm [25]

At CRYPTO 2009, Heninger and Shacham utilized the algebraic relationship between the private components to develop a branch-and-prune technique for key reconstruction with known bits. They discussed this algorithm in the context of cold boot attacks and demonstrated a lower bound on the number of known bits required for the algorithm to succeed within polynomial-time.

Let (N, e) be an RSA public key, where N is $2n$-bit and $e = 65,537$. The corresponding CRT private key set is $(p, q, d, d_p, d_q, q_{inv} = q^{-1} \mod p)$, which simultaneously satisfies the following equations: (1) $N = pq$, (2) $ed = 1 + k(p -$

$1)(q-1)$, (3) $ed_p = 1 + k_p(p-1)$ and (4) $ed_q = 1 + k_q(q-1)$, where k, k_p and k_q can be enumerated if e is small. The key exhibits high levels of redundancy, as knowledge of any of its components can lead to the factorization of N. Heninger-Shacham's algorithm leverages these constraints to progressively determine the parameters. It starts from the least significant bit, and at each iteration, partial solutions are branched or pruned depending on the knowledge of bits for that position. Specifically, given the knowledge of bits 0 to $i-1$, the i-th bit of each argument can be solved by lifting the solution from the constraint equations mod 2^i to the solution mod 2^{i+1}.

Cold boot attacks could result in partial information of $(p, q, d, d_p, d_q, q_{inv})$ being obtained at random. However, in other side-channel models, only some of them may be obtained, such as $\{p, q\}$, $\{d\}$, or $\{d_p, d_q\}$. This study focuses on the first case in which partial knowledge of p and q is revealed. The following part reviews the behavior of the algorithm in this case.

According to Hensel's lifting lemma, the dependency between the i-th bit and known bits is expressed as

$$p[i] + q[i] \equiv (N - \bar{p}\bar{q})[i] \mod 2, \tag{1}$$

where $\bar{p}$ and $\bar{q}$ denote the partial solution of p and q up to i of the bits. Given $\bar{p}$ and $\bar{q}$, the algorithm's branching behavior at bit i can be summarized as follows.

- When both $p[i]$ and $q[i]$ are unknown, the algorithm branches;
- When one of $p[i]$ and $q[i]$ is unknown, there is a unique solution;
- When the i-th bit of p and q are known, the wrong solution is pruned with approximate 50% probability (empirically).

Understanding these cases is straightforward, as the right side of equation (1) is given.

Heninger and Shacham conducted a complexity analysis of the reconstruction by examining the algorithm's local and global branching behavior. They evaluated the number of erroneous solutions generated during the process. Specifically, at bit i, the algorithm generates one correct solution and some incorrect solutions. As the number of unknown bits increases, the expansion of branches tends to be exponential; conversely, it converges swiftly when the number of unknown bits decreases.

In their random model, it is assumed that the knowledge of $p[i]$ and $q[i]$ is mutually independent, each occurring with a probability of δ. To establish the relation between the number of branches examined during the reconstruction and δ, the authors calculated the expected number of wrong solutions generated from a correct and incorrect one, denoted as $\mathrm{E}(Z_g)$ and $\mathrm{E}(W_b)$, respectively. Subsequently, using PGF and the sequence, they derived the expected total number of branches produced at step i, represented as $\mathrm{E}(X_i) = \frac{\mathrm{E}(Z_g)}{1-\mathrm{E}(W_b)}(1 - \mathrm{E}(W_b)^i)$. Given that the expressions for the expected values of $\mathrm{E}(Z_g)$ and $\mathrm{E}(W_b)$ are dependent solely on δ, it follows that when $\mathrm{E}(W_b) < 1$, $\mathrm{E}(X_i)$ can be bounded by a constant that only depends on δ.

Adding up the incorrect solutions generated at each bit, one can calculate the expected total number of branches examined for an n-bit key. Heninger and Shacham also provided the variance of X_i as a measure of how far $\sum_{i=0}^{n-1} X_i$ is spread from its average value. The following lemma plays an important role in this computation.

Lemma 1.

$$\mathrm{Var}(\sum_{i=1}^{n} X_i) \leq n^2 \max_i \mathrm{Var}(X_i)$$

In sum, if the algorithm has partial knowledge of p and q, it checks no more than $29n^2 + 29n$ keys with a probability higher than $\frac{n^2-1}{n^2}$ when $\delta = 0.59$.

3 A Novel Leakage Model

This section presents a leakage model derived from OpenSSL's implementation of the Miller–Rabin primality test. We begin by identifying the source of this leakage, followed by a formal characterization of the model and its corresponding leakage scenarios. Finally, we compare these scenarios and analyze the reduction in computational complexity observed in specific cases.

3.1 Root Cause of the Leakage Model

Assuming that an adversary is capable of extracting b bits from each window in a fixed-window modular exponentiation. The proposed leakage model, which emerges when an adversary targets the modular exponentiation invoked during the Miller-Rabin primality testing, arises for two reasons:

- The input to the modular exponentiation in the Miller-Rabin primality test is $\frac{p-1}{\mathrm{T}(p-1)}$, as shown in line 3 of Algorithm 1, where $\mathrm{T}(p-1)$ denotes the largest power of 2 that divides by $p-1$;
- At the beginning of the modular exponentiation, OpenSSL pads the exponent to the public size of p, which is n, as shown in listing 1.

The most significant 0 bits of the exponent could be disclosed by analyzing the number of iterations performed in the modular exponentiation. To avoid such a leakage, OpenSSL uses all bits stored in the exponent variable. Listing 1 is a code snippet of constant-time modular exponentiation used in OpenSSL. It computes $\mathbf{rr} = \mathbf{a}^{\mathbf{p}} \mod \mathbf{m}$. **BIGNUM** is a structure that holds a single large integer. The member variable **top** represents the number of words used. **BN_BITS2** indicates the word size specified in the *bn.h* file. **bits** corresponds to the public size of the exponent, which ensures that the number of iterations in fixed-window modular exponentiation is independent of the actual size of the exponent.

```
int bn_mod_exp_mont_fixed_top(BIGNUM *rr, const BIGNUM *a
    , const BIGNUM *p, const BIGNUM *m, BN_CTX *ctx,
    BN_MONT_CTX *in_mont)
{
    ...
    /* Use all bits stored in stored in variable p to
    prevent leakage of leading zero bits. */
    bits = p->top * BN_BITS2;
    ...
}
```

Listing 1: OpenSSL pads the exponent to its public size.

Therefore, the operands for the the modular exponentiation in OpenSSL's Miller-Rabin primality test are $p' = \frac{p-1}{\mathrm{T}(p-1)}$ or $q' = \frac{q-1}{\mathrm{T}(q-1)}$, padded to a length of n, as shown in Figure 1. In our assumption, an adversary can extract b bits from each window of these operands. Given that Heninger and Shacham's branch-and-prune algorithm works by exploiting dependencies between the i-th bits of p and q, the partial bits obtained here are not directly usable as input. A new model is required to characterize the mapping from p' and q' back to p and q, as illustrated in Figure 1.

3.2 The Proposed Leakage Model

OpenSSL implements the fixed-window exponentiation from left to right, starting from the most significant bit. Since n may not be a multiple of w, the first window can be partial, with a length of $n \bmod w$. All remaining windows are of the full w-bit size. Given this, the bits recovered under our assumption can be expressed as: $p'[xw + y]$ and $q'[xw + y]$, where $0 \le x \le \lfloor \frac{n}{w} \rfloor$, $0 \le y \le b - 1$, and $0 \le xw + y \le n - 1$. p' and q' are recovered in the same manner; thus, the i-th bits of p' and q' are known, or neither is known.

Our model describes the leaked bits as following:

$$\begin{cases} p[(xw + y + \mathrm{T}(p-1)) \bmod n], \\ q[(xw + y + \mathrm{T}(q-1)) \bmod n], \end{cases}$$

where $0 \le x \le \lfloor \frac{n}{w} \rfloor$, $0 \le y \le b-1$, $0 \le xw+y \le n-1$, and $1 \le \mathrm{T}(p-1), \mathrm{T}(q-1) \le n$. Since $\mathrm{T}(p-1)$ is not necessarily equal to $\mathrm{T}(q-1)$, the i-th bits of p and q exhibit three possible states:

- $p[i]$ and $q[i]$ are known;
- $p[i]$ and $q[i]$ are unknown;
- One of $p[i]$ and $q[i]$ is known.

Suppose $\mathrm{T}(p-1) > \mathrm{T}(q-1)$, both p and q can be partitioned into three blocks, as shown in Figure 2. Specifically,

- **Block 1** ($\mathrm{T}(q-1) > i \ge 0$): $p[i] = q[i] = 0$, except $p[0] = q[0] = 1$;

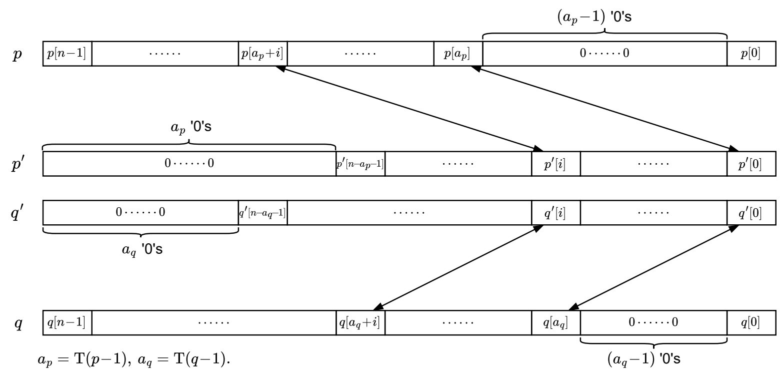


Fig. 1: Map the recovered $p'[i]$ and $q'[i]$ to p and q.

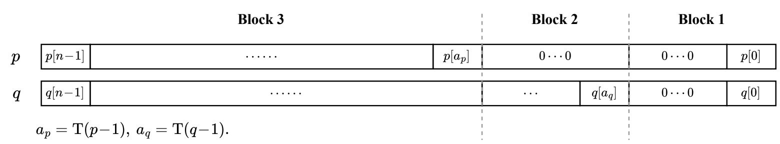


Fig. 2: p and q recovered from the side-channel oracle.

– **Block 2** ($\mathrm{T}(p-1) > i \geq \mathrm{T}(q-1)$): $p[i] = 0$ and $q[i]$ may be revealed;
– **Block 3** ($n-1 \geq i \geq \mathrm{T}(p-1)$): each window of p (or q) is revealed in a fixed pattern.

When $\mathrm{T}(p-1) < \mathrm{T}(q-1)$, the situation is analogous. **Block 2** is absent when $\mathrm{T}(p-1) = \mathrm{T}(q-1)$.

Given $\mathrm{T}(p-1)$ and $\mathrm{T}(q-1)$, **Block 1** is fully determined, and either p or q in **Block 2** can be uniquely computed, yielding exactly one valid partial solution for **Block 3**. Consequently, the efficiency of key reconstruction depends on its performance in **Block 3**, subject to the knowledge of $p[i]$ and $q[i]$ within each window. Notably, the bits recovered in **Block 3** exhibit a periodic pattern within each window. This periodic structure allows τ to be interpreted as defining a ring over the integer w, where τ is defined as the absolute difference between $\mathrm{T}(p-1)$ and $\mathrm{T}(q-1)$.

In our leakage model, **Block 3** falls into one of following scenarios based on the value of τ.

– The aligned, where $\tau \equiv 0 \bmod w$: $p[i]$ and $q[i]$ are either recovered or unknown;
– The misaligned, where $\tau \not\equiv 0 \bmod w$: $p[i]$ and $q[i]$ could be one of the following: both recovered, one recovered, or both unknown.

For simplicity, the modulus in the above equivalence will be omitted in subsequent discussions.

Table 1 demonstrates both scenarios for a window size of 6, with 4 bits successfully recovered in each window, where $*$ denotes the unknown bit. The remainder of this paper will utilize this assumption as a case study and present corresponding experimental validations. Our model is not constrained by this specific assumption. In Section 4, we develop a generalized analysis that extends our results to various assumptions, including 3-bit recovery from 5-bit windows[3] and 2-bit recovery from 4-bit windows[4].

Table 1: Left: an aligned case, Right: a misaligned case.

$i+2$	$i+1$	i	$i-1$	$i-2$	$i-3$
$*$	$*$	1	0	1	0
$*$	$*$	1	0	1	1

$i+2$	$i+1$	i	$i-1$	$i-2$	$i-3$
$*$	$*$	1	0	1	0
1	0	1	1	$*$	$*$

3.3 Reduced Complexity

The occurrence of either scenario during an attack depends on the value of τ, or more precisely, on the randomly generated primes p and q. Before analyzing the probability of these two cases, we discuss their differences.

Due to the branching behavior of the branch-and-prune algorithm at the i-th bit, under the constraint $N = pq$, knowing either $p[i]$ or $q[i]$ is sufficient to uniquely determine the other. This explains why: when $p[i]$ and $q[i]$ are unknown, only two branches are generated, not four; when $p[i]$ and $q[i]$ are known, the redundant information helps to identify and prune incorrect branches.

In the aligned scenario, $p[i]$ and $q[i]$ are unknown or known, indicating that the algorithm branches or performs pruning with a certain probability[5]. In misaligned scenarios, the frequency of such fully known or fully unknown bit-pairs decreases, and cases where only one of them is known become more common. As a result, the reconstruction involves fewer branching operations, which reduces the number of false candidates and thus lowers computational complexity. In the specific scenario studied in this paper, where at least one $p[i]$ or $q[i]$ is available at each bit, the key is reconstructed without generating false candidates.

4 Algorithm Runtime Analysis

This section explores the complexity of key reconstruction under different leakage scenarios by evaluating the number of partial solutions.

[3] This leakage pattern has been demonstrated in [51].
[4] This leakage pattern represents the specific case we analyzed for 1024-bit RSA.
[5] This probability is estimated by generating multiple pairs of p and q for complexity analysis; however, for fixed p, q, and given bits, the execution of the branch-and-prune algorithm is deterministic.

4.1 Global Branching Behavior

Given that the proposed leakage scenarios differ only in their window-level recovery patterns, we first establish a general equation characterizing the algorithm's global behavior, then investigate the local branching process in the window.

Let $Y_\imath$ denote the number of partial solutions checked at window $\imath$ in **Block 3**, and a sum of $Y_\imath$ yields the total number of partial keys examined in the third block, where $1 \leq \imath \leq \lceil (n - max(\mathrm{T}(p-1), \mathrm{T}(q-1)))/w \rceil$.

$$Y_\imath = \sum_{\jmath=1}^{X_{\imath-1}} Z_\jmath + Z_c, \tag{2}$$

where Z and Z_c are random variables that count, within a window, the number of key validation attempts triggered by a wrong and a correct solution, respectively. There is only one correct partial key in the whole process. Since the number of incorrect solutions is not constant, it is imperative to estimate how many incorrect solutions are likely to remain after lifting the window $\imath$, denoted as $X_\imath$.

Let C and W be two random variables representing the number of incorrect candidates remaining after a correct and a wrong solution passes a window in **Block 3**, respectively. Then

$$X_\imath = \sum_{\jmath=1}^{X_{\imath-1}} W_\jmath + C. \tag{3}$$

To explore the characteristics of the distribution of $Y_\imath$, we have the following expressions.

Theorem 3.

$$\mathrm{E}(Y_\imath) = \frac{\mathrm{E}(C)\mathrm{E}(Z)}{1 - \mathrm{E}(W)}(1 - \mathrm{E}(W)^{\imath-1}) + \mathrm{E}(Z_c). \tag{4}$$

Proof. See Appendix A in the full version [20].

□

Theorem 4.

$$\mathrm{Var}(Y_\imath) = c_3 \mathrm{E}(W)^{2(\imath-1)} - c_2 \mathrm{E}(W)^{\imath-1} + c_1, \tag{5}$$

with

$$\begin{aligned} c_1 &= \frac{\mathrm{E}(C)\mathrm{Var}(W) + (1 - \mathrm{E}(W))\mathrm{Var}(C)}{(1 - \mathrm{E}(W)^2)(1 - \mathrm{E}(W))}\mathrm{E}(Z)^2 \\ &+ \frac{\mathrm{E}(C)}{1 - \mathrm{E}(W)}\mathrm{Var}(Z) + \mathrm{Var}(Z_c), \\ c_2 &= \frac{\mathrm{E}(C)\ \mathrm{Var}(W)}{(1 - \mathrm{E}(W))^2\mathrm{E}(W)}\mathrm{E}(Z)^2 + \frac{\mathrm{E}(C)}{1 - \mathrm{E}(W)}\mathrm{Var}(Z), \\ c_3 &= (\frac{\mathrm{E}(C)\ \mathrm{Var}(W)}{(1 - \mathrm{E}(W)^2)(1 - \mathrm{E}(W))\mathrm{E}(W)} - \frac{\mathrm{Var}(C)}{1 - \mathrm{E}(W)^2})\mathrm{E}(Z)^2. \end{aligned}$$

The expression presented above is derived under the assumption that $\mathrm{E}(W) \neq 1$. For the sake of generality, when $\mathrm{E}(W) = 1$,

$$\mathrm{E}(Y_\imath) = (\imath - 1)\,\mathrm{E}(C)\mathrm{E}(Z) + \mathrm{E}(Z_c). \tag{6}$$

Similarly,

$$\begin{aligned}\mathrm{Var}(Y_\imath) = (&\frac{(\imath-1)(\imath-2)}{2}\mathrm{E}(C)\,\mathrm{Var}(W) + (\imath-1)\mathrm{Var}(C))\mathrm{E}(Z)^2 \\ &+ (\imath-1)\,\mathrm{E}(C)\,\mathrm{Var}(Z) + \mathrm{Var}(Z_c).\end{aligned} \tag{7}$$

Proof. See Appendix A in the full version [20].

□

4.2 Local Branching Behavior at Each Window

Recall that the program's local behavior at each bit depends on the knowledge of $p[i]$ and $q[i]$. Within a window, the leakage pattern is fixed by providing w, b, and τ, where w is given by the implementation of modular exponentiation, b is subject to the attacker's capabilities, and τ is determined by randomly generated p and q. In the reconstruction process, the only variable is the probability of a wrong branch that will be pruned when $p[i]$ and $q[i]$ are known, denoted as φ.

We consider the scenario where the least significant b bits of each window are recovered. When $\tau \equiv 0$, a correct partial solution passes the window and generates $2^{w-b} - 1$ wrong candidates. An incorrect partial solution produces 2^{w-b} erroneous candidates after passing the evaluation with probability $(1-\varphi)^b$. This is the case of alignment.

Depending on the degree of misalignment, the number of situations where $p[i]$ and $q[i]$ are unknown within a window, denoted as Δ, could be reduced at most to

$$\begin{cases} 0 & \text{if } w - b \leq b, \\ w - 2b & \text{if } w - b > b. \end{cases}$$

Considering the attack on fixed-window modular exponentiation, $w - b > b$ implies that less than half of the bits of p and q are obtained. This situation produces a large number of incorrect candidates, rendering key reconstruction infeasible within a reasonable time.[6] Hence, only the case of $w-b \leq b$ is discussed in this study.

Assuming $\mathrm{T}(p-1) \geq \mathrm{T}(q-1)$, $\overbrace{*...*}^{w-b}\ \overbrace{\diamond...\diamond}^{b}$ denotes the least window of p in **Block 3**, where $*$ and $\diamond$ represent unknown and known bits, respectively.[7] As τ increases from 0 to w, the leakage pattern in the corresponding window of q constructs a cycle, while Δ initially decreases from $w-b$ to 0 and then increases back to $w-b$, as shown in Figure 3.

[6] According to [42], the branch-and-prune algorithm works if more than 50% of the secret bits are acquired. Although known bits could be slightly lower than 50%, such as $p[i]$ and $q[i]$ are recovered alternatively, $w - b > b$ indicates two many erasures.

[7] $\mathrm{T}(p-1) < \mathrm{T}(q-1)$ is analogous since p and q are equivalent here.

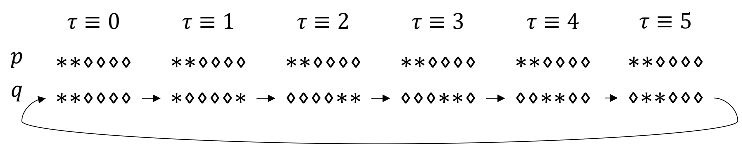


Fig. 3: Take $w = 6$ and $b = 4$ as an example to illustrate the leakage pattern of the window in **Block 3**.

Based on the branching behavior exhibited by the algorithm, it can be observed that C, the number of incorrect candidates remaining after a correct solution passes a window in **Block 3**, is directly related to Δ.

$$\mathrm{E}(C) = 2^{\Delta} - 1, \qquad \mathrm{E}(C^2) = (2^{\Delta} - 1)^2.$$

Let Λ denote the number of bits where $p[i]$ and $q[i]$ are known in the window. Λ determines the probability of retaining an incorrect input solution, Δ specifies how this solution forks.

$$\mathrm{E}(W) = 2^{\Delta}(1-\varphi)^{\Lambda}, \qquad \mathrm{E}(W^2) = 2^{2\Delta}(1-\varphi)^{\Lambda}.$$

When discussing the number of partial keys verified within a given window, it should be noted that Z and Z_c are determined by Δ, Λ, and the position of known bits. Consider $\tau \equiv 1$ and $\tau \equiv 5$ depicted in Figure 3. When $\tau \equiv 1$, a correct partial key passes through five bits and branches at the last bit. However, the latter case produces a false branch at the 5-th bit, requiring an additional calculation at the 6-th bit.

The general expressions for the expected values and variances of Z and Z_c require knowledge of the position of known bits, leading to complex expressions. To simplify the computation, we provide $\mathrm{E}(Z_c)$, $\mathrm{E}(Z_c^2)$, $\mathrm{E}(Z)$, and $\mathrm{E}(Z^2)$ under specific w and b, as shown in Table 2. These results can be directly extended to other scenarios.

To demonstrate the results in Table 2, we use the alignment case as an example. After the first 4 evaluations, a correct solution passes the lower 4 bits of the window. Two branches are generated after evaluating the next bit. Then, two candidates are evaluated at the last bit. Thus, a correct solution involves a total of 7 evaluations. In contrast, an incorrect solution may be discarded after each of the first 4 evaluations with a probability of φ. Once an incorrect solution is retained with probability $(1-\varphi)^4$, it behaves identically to a correct solution in the last two bits. Consequently, $\mathrm{E}(Z) = \sum_{i=1}^{4} i(1-\varphi)^{i-1}\varphi + 7(1-\varphi)^4$.

Under the random model, [25] empirically demonstrates that the probability of independently pruning an incorrect candidate approaches $1/2$ for known $p[i]$ and $q[i]$. Since our scenarios can be viewed as special cases of this random model, $\varphi = 1/2$ is expected to hold here. Our simulations confirm this expectation.

Table 2: The expected values and variances for Z and Z_c when $w = 6$ and $b = 4$.

	$\tau \equiv 0$	$\tau \equiv 1$	$\tau \equiv 2$	$\tau \equiv 3$	$\tau \equiv 4$	$\tau \equiv 5$
$\mathrm{E}(Z_c)$	7	6	6	6	6	7
$\mathrm{E}(Z_c^2)$	49	36	36	36	36	49
$\mathrm{E}(Z)$	$\frac{33}{16}$	3	4	3	$\frac{5}{2}$	$\frac{9}{4}$
$\mathrm{E}(Z^2)$	$\frac{107}{16}$	$\frac{43}{4}$	$\frac{35}{2}$	$\frac{27}{2}$	$\frac{21}{2}$	$\frac{35}{4}$

4.3 Bounding the Total Number of Keys Examined

The total number of keys examined over an entire program can be represented as

$$Y = |\mathrm{T}(p-1) - \mathrm{T}(q-1)| + \sum_{\imath=1}^{l} Y_\imath + \varepsilon,$$

where $l = \lfloor (n - max(\mathrm{T}(p-1), \mathrm{T}(q-1)))/w \rfloor$ is the number of complete windows in **Block 3**. Specifically, the first term counts the evaluations executed in blocks 2. The second term plus ε represents the total number of keys examined in **Block 3**, where ε denotes the number of keys evaluated in the last window. If $n - max(\mathrm{T}(p-1), \mathrm{T}(q-1))$ is divisible by w, $\varepsilon = 0$. For $\mathrm{E}(W) \le 1$, the program either converges continuously or alternates between divergence and convergence in each window. Thus, the tree does not spread exponentially, which implies that ε will not be too large. In addition, empirical investigations demonstrate that $\mathrm{T}(p-1)$ and $\mathrm{T}(q-1)$ are usually small, as shown in Figure 4. Therefore, the primary contribution to $\mathrm{E}(Y)$ arises from the accumulation of $\mathrm{E}(Y_\imath)$.

Given w, b, τ, and estimation of φ, $\mathrm{E}(C)$, $\mathrm{E}(W)$, $\mathrm{E}(Z_c)$, and $\mathrm{E}(Z)$ are constants. According to Theorem 3, we give a bound on $\mathrm{E}(Y)$ to facilitate estimation.

$$\mathrm{E}(Y) < \mathrm{E}\left(\sum_{\imath=1}^{\lceil \frac{n}{w} \rceil} Y_\imath\right) \tag{8}$$

$$\begin{cases} \le \left\lceil \dfrac{n}{w} \right\rceil \left(\dfrac{\mathrm{E}(C)\mathrm{E}(Z)}{1-\mathrm{E}(W)} + \mathrm{E}(Z_c) \right) & \text{if } \mathrm{E}(W) < 1, \\ = \dfrac{\lceil \frac{n}{w} \rceil^2 - \lceil \frac{n}{w} \rceil}{2} \mathrm{E}(C)\,\mathrm{E}(Z) + \left\lceil \dfrac{n}{w} \right\rceil \mathrm{E}(Z_c) & \text{if } \mathrm{E}(W) = 1. \end{cases}$$

Next, Lemma 1 gives a bound for the variance of $\sum_\imath Y_\imath$, where $1 \le \imath \le \lceil \frac{n}{w} \rceil$.

$$\mathrm{Var}(\sum_\imath Y_\imath) \le \left\lceil \frac{n}{w} \right\rceil^2 \max_\imath \mathrm{Var}(Y_\imath).$$

From equation 5, we observe that it converges to c_1 as $\imath$ increases. When $\imath = 1$, $c_3 - c_2 \le 0$. As $\imath$ grows, the expression $c_3 \mathrm{E}(W)^{2(\imath-1)} - c_2 \mathrm{E}(W)^{\imath-1}$ remains negative but increases monotonically. Consequently, $\max \mathrm{Var}(Y_\imath) = c_1$ at $\imath =$

$\lceil \frac{n}{w} \rceil$, where c_1 is a constant determined by w, b, τ, and the estimate of φ. The result holds when $\mathrm{E}(W) = 1$, where the maximum variance occurs at $\imath = \lceil \frac{n}{w} \rceil$.

Table 3 gives $\mathrm{E}(Y)$ and $\mathrm{Var}(Y)$ when $w = 6$, $b = 4$, and $\varphi = 1/2$. Our analysis reveals that both: (i) the expected number of keys examined during reconstruction, and (ii) their degree of dispersion are lower in misaligned cases compared to aligned cases. Despite the fact that the number of recovered bits remains constant, the system exhibits significantly reduced complexity in misalignment scenarios. Furthermore, the cases in which $\tau \equiv 2, 3, \text{and} 4$ yield identical estimates for $\mathrm{E}(Y)$ and $\mathrm{Var}(Y)$. These cases can be treated as equivalent in subsequent analyses. Although cases where $\tau \equiv 2$ and $\tau \equiv 3$ show minor differences in $\mathrm{Var}(Y)$, we classify them into the same category. This grouping is justified because both have exactly one bit where $p[i]$ and $q[i]$ are recovered. The observed difference in variance arises from the position of this bit.

Table 3: $\mathrm{E}(Y)$ and $\mathrm{Var}(Y)$ when $w = 6$, $b = 4$, and $\varphi = 1/2$.

	$\tau \equiv 0$	$\tau \equiv 1$	$\tau \equiv 2$	$\tau \equiv 3$	$\tau \equiv 4$	$\tau \equiv 5$
$\mathrm{E}(C)$	3	1	0	0	0	1
$\mathrm{E}(W)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\mathrm{E}(Z_c)$	7	6	6	6	6	7
$\mathrm{E}(Z)$	$\frac{33}{16}$	3	4	3	$\frac{5}{2}$	$\frac{9}{4}$
$\mathrm{E}(Y) <$	$\frac{61}{4}\lceil \frac{n}{6} \rceil$	$10\lceil \frac{n}{6} \rceil$	$6\lceil \frac{n}{6} \rceil$	$6\lceil \frac{n}{6} \rceil$	$6\lceil \frac{n}{6} \rceil$	$10\lceil \frac{n}{6} \rceil$
$\mathrm{Var}(C)$	0	0	0	0	0	0
$\mathrm{Var}(W)$	$\frac{15}{16}$	$\frac{7}{16}$	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{7}{16}$
$\mathrm{Var}(Z_c)$	0	0	0	0	0	0
$\mathrm{Var}(Z)$	$\frac{623}{256}$	$\frac{7}{4}$	$\frac{3}{2}$	$\frac{9}{2}$	$\frac{17}{4}$	$\frac{59}{16}$
$\mathrm{Var}(Y) \leq$	$\frac{107}{4}\lceil \frac{n}{6} \rceil^2$	$\frac{119}{15}\lceil \frac{n}{6} \rceil^2$	0	0	0	$\frac{121}{15}\lceil \frac{n}{6} \rceil^2$

4.4 Discussion on Additional Erasures

Results in Table 3 suggest that the algorithm could be reconstructed in a reasonable time, even if the recovered partial key contains additional erasures. In this subsection, we examine scenarios in which some windows are totally erased. This happens in some SCAs. For example, in cache-timing attacks, system interruptions may lead to information loss [7, 50]. In [38], system noise in power or EM traces may cause some windows to be recovered with a low probability.

Let ρ be the probability that a given window fails to be recovered in either p or q. In this context, the recovery of a window does not imply obtaining all bits, but rather refers specifically to the recovery corresponding to our assumption. For each window, there exist three possible cases:

- Both of p and q are recovered with probability $(1-\rho)^2$;
- One of p and q is recovered with probability $2\rho(1-\rho)$;

– Both of p and q fail to be recovered with probability ρ^2.

As additional erasures do not impact the algorithm's global behavior, we only need to re-estimate C, W, Z_c and Z.

$$\begin{aligned} \mathrm{E}(C) &= (2^{\Delta} - 1)(1-\rho)^2 + (2^{w-b} - 1)(2\rho - 2\rho^2) \\ &\quad + (2^{w} - 1)\rho^2, \\ \mathrm{E}(W) &= 2^{\Delta}(1-\varphi)^{\Lambda}(1-\rho)^2 + 2^{w-b}(2\rho - 2\rho^2) + 2^{w}\rho^2. \end{aligned} \tag{9}$$

According to equation (8), $\mathrm{E}(W)$ should be ≤ 1 to avoid an exponential increase in the complexity of the key reconstruction. Given w, b, φ, the expression for $\mathrm{E}(W)$ in (9) depends solely on ρ [8].

Case1: $w = 6$, $b = 4$, and $\varphi = 1/2$.

$$\mathrm{E}(W) = \frac{225}{4}\rho^2 + \frac{15}{2}\rho + \frac{1}{4}.$$

Solving the inequality $\mathrm{E}(W) \leq 1$ yields $-\frac{1}{5} \leq \rho \leq \frac{1}{15}$. Since ρ is a probability that should be in $[0, 1]$, we have $\rho \leq \frac{1}{15}$.

Table 4: $\mathrm{E}(Y)$ and $\mathrm{Var}(Y)$ when $w = 6$, $b = 4$, $\varphi = 1/2$ and $\rho = \frac{1}{15}$.

	$\tau \equiv 0$	$\tau \equiv 1$	$\tau \equiv 2$	$\tau \equiv 3$	$\tau \equiv 4$	$\tau \equiv 5$
$\mathrm{E}(C)$	$\frac{49}{15}$	$\frac{343}{225}$	$\frac{49}{75}$	$\frac{49}{75}$	$\frac{49}{75}$	$\frac{343}{225}$
$\mathrm{E}(W)$	1	1	1	1	1	1
$\mathrm{E}(Z_c)$	$\frac{1{,}631}{225}$	$\frac{497}{75}$	$\frac{1{,}603}{225}$	$\frac{1{,}561}{225}$	$\frac{1{,}519}{225}$	$\frac{1{,}673}{225}$
$\mathrm{E}(Z)$	$\frac{2{,}653}{900}$	$\frac{301}{75}$	$\frac{1{,}211}{225}$	$\frac{973}{225}$	$\frac{833}{225}$	$\frac{742}{225}$
$\mathrm{E}(Y) \approx$	$4.8\left\lceil\frac{n}{6}\right\rceil^2$	$3.1\left\lceil\frac{n}{6}\right\rceil^2$	$1.8\left\lceil\frac{n}{6}\right\rceil^2$	$1.4\left\lceil\frac{n}{6}\right\rceil^2$	$1.2\left\lceil\frac{n}{6}\right\rceil^2$	$2.5\left\lceil\frac{n}{6}\right\rceil^2$
$\mathrm{Var}(C)$	$\frac{3{,}584}{225}$	$\frac{876{,}176}{50{,}625}$	$\frac{103{,}124}{5{,}625}$	$\frac{103{,}124}{5{,}625}$	$\frac{103{,}124}{5{,}625}$	$\frac{876{,}176}{50{,}625}$
$\mathrm{Var}(W)$	$\frac{301}{15}$	$\frac{4{,}417}{225}$	$\frac{1{,}456}{75}$	$\frac{1{,}456}{75}$	$\frac{1{,}456}{75}$	$\frac{4{,}417}{225}$
$\mathrm{Var}(Z_c)$	$\frac{702{,}464}{50{,}625}$	$\frac{29{,}372}{1{,}875}$	$\frac{1{,}202{,}616}{50{,}625}$	$\frac{1{,}004{,}654}{50{,}625}$	$\frac{57{,}736}{3{,}375}$	$\frac{724{,}346}{50{,}625}$
$\mathrm{Var}(Z)$	$\frac{16{,}907{,}891}{8{,}100{,}000}$	$\frac{120{,}799}{5{,}625}$	$\frac{1{,}489{,}754}{50{,}625}$	$\frac{1{,}502{,}396}{50{,}625}$	$\frac{1{,}348{,}886}{50{,}625}$	$\frac{1{,}197{,}686}{50{,}625}$
$\mathrm{Var}(Y) \approx$	$47.5\left\lceil\frac{n}{5}\right\rceil^3$	$40\left\lceil\frac{n}{5}\right\rceil^3$	$30.6\left\lceil\frac{n}{5}\right\rceil^3$	$19.8\left\lceil\frac{n}{5}\right\rceil^3$	$14.5\left\lceil\frac{n}{5}\right\rceil^3$	$11.5\left\lceil\frac{n}{5}\right\rceil^3$

Table 4 presents the estimation of $\mathrm{E}(Y)$ and $\mathrm{Var}(Y)$ computed from equations (6) and (7), accounting for $\frac{1}{15}$% unrecovered windows (equivalent to additional erasures). Here, $\approx$ is used instead of $<$ and $\leq$, as we discard lower-order polynomial terms that have a negligible impact on overall expression. Comparing Table 4 with Table 3, the expected number of partial keys verified in the reconstruction scales quadratically with n (as opposed to linearly), due to the introduced erasures. It is observed that the speed advantage in misaligned cases persists even when the introduced erasures approach the algorithm's maximum tolerance. Moreover, the variance of Y indicates that the aligned case is more prone to generating candidates with larger search spaces, which leads to a longer time for key reconstruction in practice.

[8] $\mathrm{E}(W)$ is independent of Δ and Λ if $\varphi = \frac{1}{2}$.

Case2: $w = 4$, $b = 2$, and $\varphi = 1/2$.

$$\mathrm{E}(W) = 9\rho^2 + 6\rho + 1.$$

Solving the inequality $\mathrm{E}(W) \leq 1$ yields $\rho = 0$. This is consistent with the results of [25] and [42], which show that revealing 50% of the bits of p and q is necessary to prevent the algorithm's complexity from growing exponentially.

In this case, $\frac{b}{w} = 50\%$ indicates that the key reconstruction cannot tolerate additional erasures. However, $\tau = 2$ in Table 5 suggests that allowing some additional erasures may be feasible in practice. When $\tau = 2$, the reconstruction process verifies at most n correct partial keys. In practical scenarios, although an additional 25-bit erasure would expand the candidate solution space up to 2^{25}, the computational cost remains well within the means of modestly resourced adversaries [16]. For 1024-bit RSA($w = 4$), an additional 25-bit erasure corresponds to approximately 10% windows remaining unrecovered.

Table 5: $\mathrm{E}(Y)$ when $w = 4$, $b = 2$, and $\varphi = 1/2$.

	$\tau \equiv 0$	$\tau \equiv 1$	$\tau \equiv 2$	$\tau \equiv 3$
$\mathrm{E}(C)$	3	1	0	1
$\mathrm{E}(W)$	1	1	1	1
$\mathrm{E}(Z_c)$	5	4	4	5
$\mathrm{E}(Z)$	$\frac{9}{4}$	3	4	3
$\mathrm{E}(Y) <$	$\frac{27}{8}\left\lceil\frac{n}{4}\right\rceil^2 + \frac{27}{8}\left\lceil\frac{n}{4}\right\rceil$	$\frac{3}{2}\left\lceil\frac{n}{4}\right\rceil^2 + \frac{5}{2}\left\lceil\frac{n}{4}\right\rceil$	$4\left\lceil\frac{n}{4}\right\rceil$	$\frac{3}{2}\left\lceil\frac{n}{4}\right\rceil^2 + \frac{7}{2}\left\lceil\frac{n}{4}\right\rceil$

In sum, misaligned cases outperform the aligned case in both efficiency and tolerance to additional erasures. Although it is possible in SCAs that the recovered partial key contains errors, we omit this discussion in this work. According to the result presented in [33], the success rate is as low as 4% when p and q contain 1% errors and 25% erasures. In the absence of more efficient error-correction algorithms, the conventional approach treats potential errors as erasures in subsequent processing.

5 Experiment Evaluation

This section illustrates the practical implications of our proposed model from the following perspectives.

- Statistical analysis of the distribution of aligned and misaligned scenarios confirms that they exhibit non-negligible occurrence probabilities.
- SPA attacks are conducted on the target implementation, successfully recovering $\mathrm{T}(x-1)$ and the lower 4 bits within each 6-bit window, empirically validating the practical implications of our leakage assumption.
- An evaluation of key reconstruction is performed between theoretical predictions and practical outcomes.

5.1 Experimental Setup

The experiments in Sections 5.2 and 5.4 were performed on an AMAX server equipped with 64 CPU cores and 503 GB of RAM. A Python module `pyOpenSSL` is used to generate the key pairs of RSA. Leveraging multiprocess parallel computing, each experiment was completed in minutes to tens of minutes.

The experiments in Section 5.3 were carried out on the ChipWhisperer CW308 platform with an STM32F303 target board. The STM32F303 features an ARM Cortex-M4 core operating at 7.38 MHz. Power traces were recorded using CW-Lite at a sampling rate of 4×7.38 MS/s (4 samples per clock cycle) with 10-bit resolution. The trace processing was performed on a ThinkPad X1 laptop running Ubuntu 20.04, equipped with an Intel i7-8550U CPU (1.80 GHz) and 8 GB of RAM. We targeted the RSA implementation in the current version of OpenSSL, using an additional trigger signal to facilitate trace recording. Our analysis focuses on two functions:

– `BN_is_bit_set()`, invoked by `ossl_bn_Miller_rabin_is_prime()` in the source file `bn_prime.c`.
– `MOD_EXP_CTIME_COPY_FROM_PREBUF()`, called by `bn_mod_exp_mont_fixed_top()` in the source file `bn_exp.c`.

5.2 The Distribution of Aligned and Misaligned Cases

To demonstrate that the misaligned cases occur with non-negligible probability, we conducted the statistical analysis by repeatedly invoking OpenSSL to generate the private key set, from which we extracted p and q.

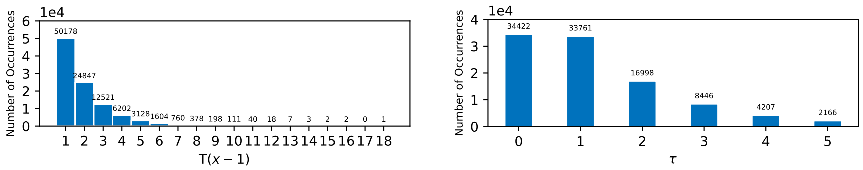
This test was performed 100,000 times for 1024-, 2048-, and 4096-bit RSA, respectively. We recorded the value of $\mathrm{T}(p-1)$ and $\mathrm{T}(q-1)$. Due to the symmetric roles of p and q in this context, the distributions of $\mathrm{T}(p-1)$ and $\mathrm{T}(q-1)$ are identical. Figure 4.(a) presents the distributions of $\mathrm{T}(x-1)$ for 2048-bit RSA, where x represents either prime. Notably, $\mathrm{T}(x-1)$ consistently takes relatively small values. Figure 4.(b) shows the distributions for the proposed scenarios. As noted previously, these scenarios can be classified by the value of Δ due to the similarity of the computational complexity. Specifically:

– Class 1 ($\tau \equiv 0$): 34,422;
– Class 2 ($\tau \equiv 1, 5$): 35,927;
– Class 3 ($\tau \equiv 2, 3, 4$): 29,651;

Each class occurs with non-negligible probability. Similar behavior is observed for both RSA-1024 and RSA-4096.

5.3 Recovering $\mathbf{T}(x-1)$ and `idx[3:0]` via SPA Attacks

Recovering $\mathbf{T}(x-1)$ To map the information recovered by SCAs back to p and q, the values of $\mathrm{T}(p-1)$ and $\mathrm{T}(q-1)$ need to be known. In OpenSSL,



(a) (b)

Fig. 4: (a) and (b) represent the distributions of $\mathrm{T}(x-1)$ and τ, respectively, for RSA-2048 over 100,000 trials.

the Miller-Rabin primality test calculates $\mathrm{T}(x-1)$ and reduces x to x' before invoking the modular exponentiation.

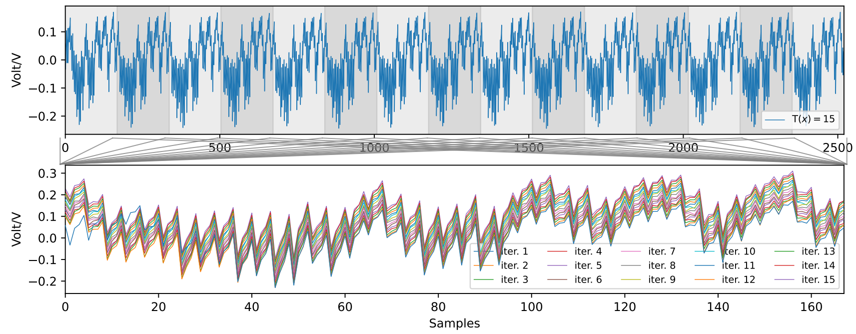
Listing 2 is a code snippet illustrating the process of calculating $\mathrm{T}(x-1)$ in OpenSSL. The implementation repeatedly calls `BN_is_bit_set()`, which tests whether the `a`-th bit in `w1` is set to 1. Since this operation is not constant-time, the exact number of function calls, which is $\mathrm{T}(x-1)$, can be leaked through the power channel.

```
1 int ossl_bn_Miller_rabin_is_prime(const BIGNUM *w, int
      iterations, BN_CTX *ctx, BN_GENCB *cb, int enhanced,
      int *status)
2 {
3     ...
4     /* (Step 1) Calculate largest integer 'a' such that
      2^a divides w-1 */
5     a = 1;
6     while (!BN_is_bit_set(w1, a))
7         a++;
8     /* (Step 2) m = (w-1) / 2^a */
9     if (!BN_rshift(m, w1, a))
10        goto err;
11    ...
12 }
```

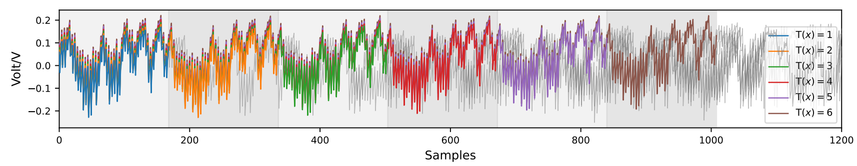
Listing 2: Computing $\mathrm{T}(x-1)$ in OpenSSL.

In function `ossl_bn_Miller_rabin_is_prime()`, the variable `w` holds the candidate prime to be tested. line 6–7 in Listing 2 sequentially check each bit of `w-1`, starting from the second least significant bit, to count the number of trailing zeros. This operation exhibits two properties that introduce a distinct timing channel exploitable via SPA: (1) The execution time of this operation depends on the value of $\mathrm{T}(x-1)$; (2) The execution time for each iteration is nearly constant.

Our experimental analysis successfully identified the power trace corresponding to a single iteration, enabling a precise determination of the iteration count through visual inspection of the power traces. Figure 5.(a) displays the power trace when $\mathrm{T}(x-1) = 15$. Each iteration is distinguishable, separated by alter-



(a)



(b)

Fig. 5: Determining the value of $\mathrm{T}(x-1)$ via SPA.

nating gray and white intervals. Although all subsequent intervals contain the 168 samples, the first interval initially contains a reduced count due to missing control instructions in the initial loop. To ensure consistent analysis, we padded the first iteration's samples to 168 by borrowing preceding data points. This preprocessing explains the visible deviation in the blue trace (Bottom of Figure 5.(a)) compared to the others. To enhance visibility, we deliberately offset the traces vertically along the y-axis, as well as in Figure 5.(b). Notably, the power trace of all subsequent iterations exhibits highly consistent patterns, enabling a reliable identification of the iteration count through visual inspection.

Figure 5.(b) presents a comparison of power traces for $\mathrm{T}(x-1) \equiv 1$ to 6. The colored traces represent the execution of the loop body, whereas the gray traces correspond to unrelated operations. These colored traces overlap at the execution of each iteration. A straightforward method to determine the number of iterations involves cross-correlating a reference single-loop trace (or a reference trace that contains 30 loops) with the target trace, and the number of overlapping segments directly yields the value of $\mathrm{T}(x-1)$.

Moreover, Figure 4 indicates $\mathrm{T}(p-1)$ and $\mathrm{T}(q-1)$ are typically small. Hence, an exhaustive search for possible values of these two parameters may also lead to successful key reconstruction.

Recovering Partial Bits of p and q In this subsection, we conducted an SPA attack on the fixed-window modular exponentiation implemented in OpenSSL to validate the practical implications of our leakage assumption.

Target function: Listing 3 presents the core implementation of fixed-window modular exponentiation in OpenSSL. The computation proceeds in each window as follows:

- Line 4-5 perform the squaring.
- Line 6-7 extract the current window of bits from the exponent.
- Line 9 retrieves the precomputed multiplier corresponding to the bits extracted in line 7.
- Line 10 multiplies the intermediate result by the retrieved multiplier.

```
1 int bn_mod_exp_mont_fixed_top(BIGNUM *rr, const BIGNUM *a
      , const BIGNUM *p, const BIGNUM *m, BN_CTX *ctx,
      BN_MONT_CTX *in_mont)
2 {
3     ...
4     for (i = 0; i < window; i++)
5         bn_mul_mont_fixed_top(&tmp, &tmp, &tmp, mont,ctx)
6     bits -= window;
7     wvalue = bn_get_bits(p, bits) & wmask;
8     /* Fetch the appropriate  precomputed value */
9     MOD_EXP_CTIME_COPY_FROM_PREBUF(&am, top, powerbuf,
      wvalue,window)
10    bn_mul_mont_fixed_top(&tmp, &tmp, &am, mont, ctx)
11    ...
12 }
```

Listing 3: fixed-window modular exponentiation in the current version of OpenSSL.

Listing 4 defines the target function `MOD_EXP_CTIME_COPY_FROM_PREBUF()`. For a window size of 6, `buf` holds 64 precomputed multipliers, and their arrangement can be visualized as a 4×16 table. The variable `idx` (or `wvalue` in Listing 3) acts as the index for selecting the multiplier to read, where its value represents the partial secret. To protect `idx` from potential attacks, this function employs a dummy load that sequentially reads all 64 precomputed multipliers and selects the correct one via the masking technique. The mask is computed as: `(0 - constant_time_eq_int()&1)`. This expression yields $\mathbf{0}$ (all zeros) or $-\mathbf{1}$ (all ones, in two's complement).

```
int MOD_EXP_CTIME_COPY_FROM_PREBUF(BIGNUM *b, int top,
    unsigned char *buf, int idx, int window)
{
    ...
    int width = 1 << window;
    int xstride = 1 << (window - 2);
    BN_ULONG y0, y1, y2, y3;
    i = idx >> (window - 2);
    idx &= xstride - 1;

    y0 = (BN_ULONG)0 - (constant_time_eq_int(i,0)&1);
```

```
11      y1 = (BN_ULONG)0 - (constant_time_eq_int(i,1)&1);
12      y2 = (BN_ULONG)0 - (constant_time_eq_int(i,2)&1);
13      y3 = (BN_ULONG)0 - (constant_time_eq_int(i,3)&1);
14
15      for (i = 0; i < top; i++, table += width) {
16          BN_ULONG acc = 0;
17          for (j = 0; j < xstride; j++) {
18                  acc |= ( (table[j + 0 * xstride] & y0) |
19                           (table[j + 1 * xstride] & y1) |
20                           (table[j + 2 * xstride] & y2) |
21                           (table[j + 3 * xstride] & y3) )
22                         & ((BN_ULONG)0 - (
        constant_time_eq_int(j,idx)&1));}
23          b->d[i] = acc;}
24      ...
25  }
```

Listing 4: Implementation of fetching the precomputed multiplier.

In line 7-8, `idx` is decomposed into: (1) the upper 2 bits (stored in `i`) as the row index; (2) the lower 4 bits (stored in `idx`) as the column index. Then line 10–13 make the selection of the row, where the mask equals to all ones only when the row index matches the upper 2 bits of `idx`. Similarly, in line 15-22, the mask outputs all ones only when the loop counter matches the lower 4 bits of `idx` and **0** otherwise. By distinguishing whether the mask is **0** or all ones, we can recover secret bits in each window. Our experiments focus on line 15-22 in Listing 4, which handle the extraction of 4 bits per window(`idx[3:0]`), rather than retrieving all bits in each 6-bit window. This is because the computation in line 10-13 is too brief to reveal detectable leaks through SPA.

The top of Figure 6 shows a trace fragment for `MOD_EXP_CTIME_COPY_FROM_PREBUF()`, extracted from the complete power trace, where `idx[3:0]`$= 0$. The target function performs 16 iterations, line 22 in Listing 4 compares `idx[3:0]` with `j`, where `j` is range from 0 to 15. `(0 - constant_time_eq_int()&1)` outputs all ones when `idx[3:0]`=`j`. We highlighted the 16 iterations in the target function using alternating gray and white backgrounds, and a distinct downward dip can be observed in the first block, indicating `idx[3:0]`$= 0$. The middle plot of Figure 6 overlays the traces of all 16 iterations, where the red line represents `idx[3:0]`$= 0$, while the remaining 15 lines are shown in gray. Zooming in (bottom plot), the red line is visually distinct from the gray lines. Similar results are observed when `idx[3:0]` takes values from 1 to 15.

Figure 7 shows the trace fragments extracted from the power trace, with each subplot representing the sensitive operations in `MOD_EXP_CTIME_COPY_FROM_PREBUF()`. The subplots are sorted by `idx[3:0]`, ranging from 0 to 15. Red arrows in the subplots mark leakage locations. It can be observed that the values of `idx[3:0]` corresponding to different traces can be directly distinguished through visual inspection. In sum, we could recover the lower 4 bits of each window by launching SPA attack on an ARM Cortex-M4 device, demonstrating power-based leakage.

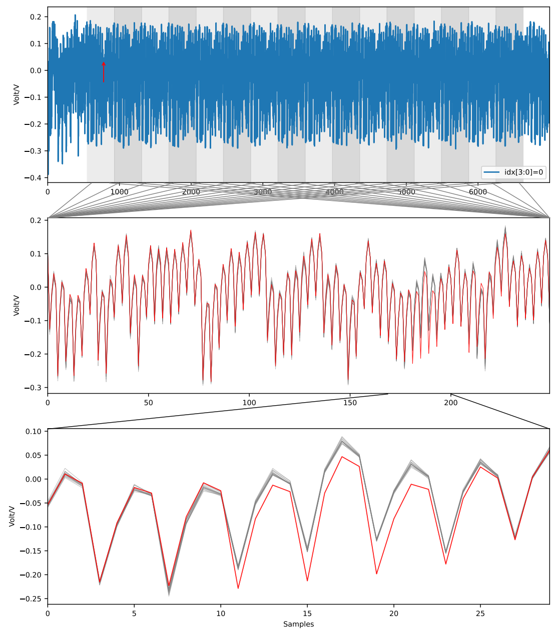


Fig. 6: The power traces of the sensitive operations in `MOD_EXP_CTIME_COPY_FROM_PREBUF()`

5.4 An Evaluation of Key Reconstruction

Figure 8 displays the distributions of incorrect partial solutions observed in 100,000 trials for RSA-2048 and RSA-4096. The green line represents cases where $\tau \equiv 2$, 3 and 4, none of which produced incorrect solutions. These scenarios occur 29,651 times for RSA-2048 and 29,545 times for RSA-4096, respectively. We adopted non-uniform intervals in this figure since the number of incorrect solutions in the remaining two classes varied significantly, ranging from hundreds to hundreds of thousands. For example, when $\tau \equiv 0$, the false solutions in Figure 8.(b) varied between $1,849$ to $111,723$. As the interval width increases, the probability that the number of incorrect solutions falls within a given range drops rapidly.[9] The blue and orange lines demonstrate that most the results cluster around the expected value, with only a few trails significantly exceeding it. This observed distribution aligns well with our theoretical predictions.

Table 6 compares the average number of incorrect partial solutions examined in theoretical estimates and experimental results. For both RSA-2048 and RSA-4096, the first row corresponds to the values derived from table 3, representing our estimated count of false branches generated during key reconstruction when

[9] Interestingly, both of the figures have an unexplained increase in the interval $[5,000-6,000]$. We skip it here because of its minimal influence on the overall analysis. The increase in $[10,000-20,000]$ is reasonable since the interval is enlarged by a factor of 10.

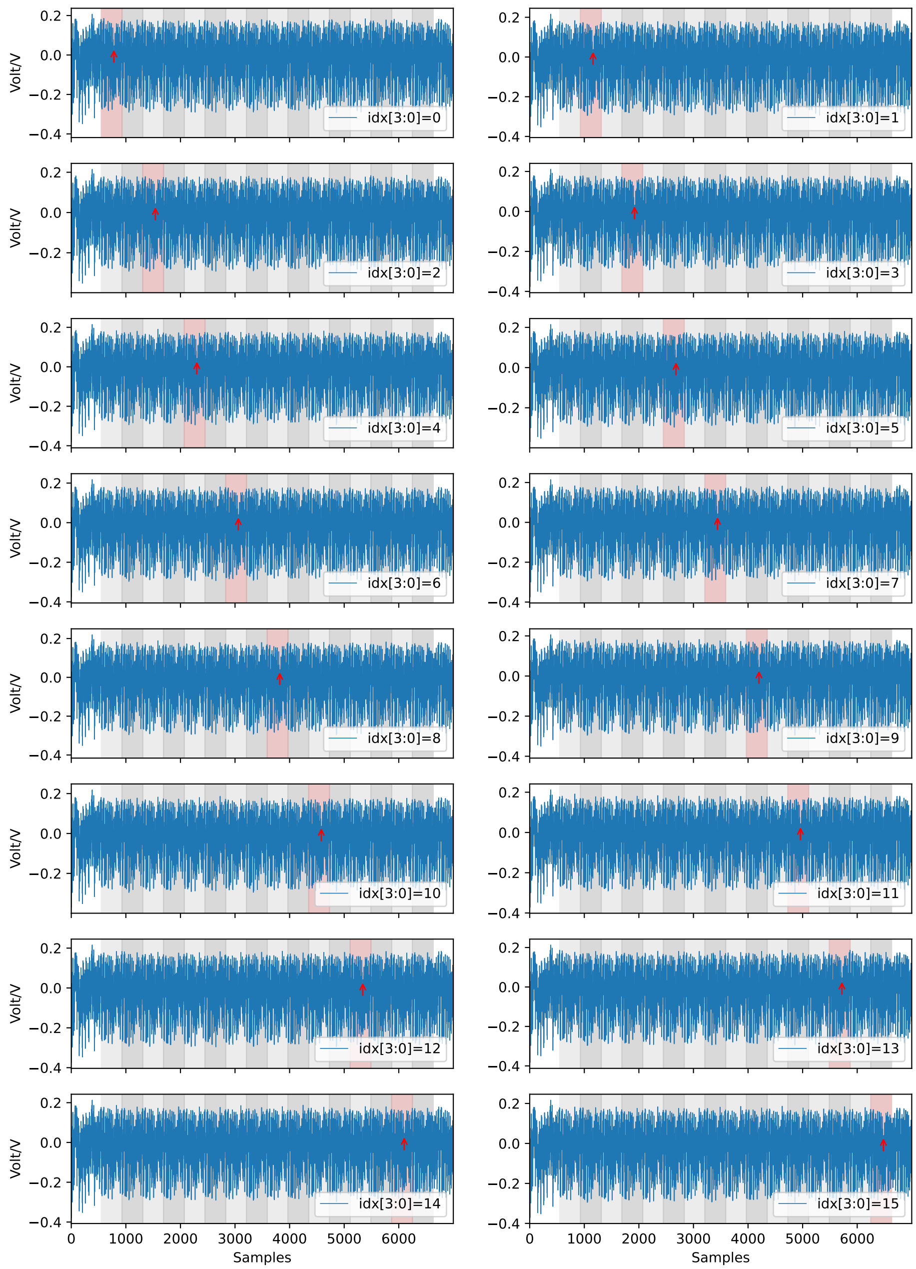
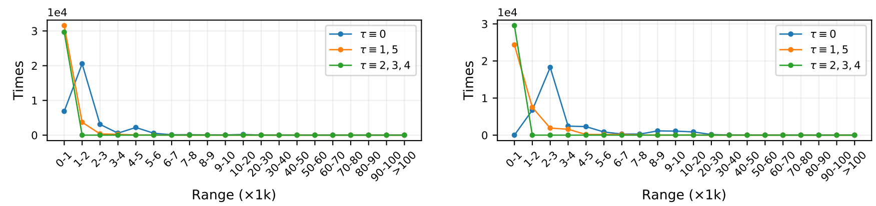


Fig. 7: The power traces of the sensitive operations in `MOD_EXP_CTIME_COPY_FROM_PREBUF()`, where `idx[3:0]` takes values from 1 to 15.



(a) RSA-2048 (b) RSA-4096

Fig. 8: The distributions of incorrect partial solutions examined in 100,000 experiments for RSA-2048 and RSA-4096.

$\varphi = 1/2$. The second row demonstrates the average number of false branches counted in our experiments. The observed discrepancies between theoretical predictions and empirical results may arise from inaccuracies in estimating φ, which implies a potential relationship between φ and the uncertainty in the key reconstruction. Furthermore, the third row presents the observed proportions of the three scenarios among 100,000 trials. In summary, the estimated complexity of the key reconstruction is consistent with results observed in practical attacks.

As shown in Table 6, the difference in average complexity between the alignment and misalignment cases seems trivial, since hundreds of false candidates can be examined in seconds on a modern computer. However, in scenarios that generate extensive candidates, especially in RSA-4096, key reconstruction may take several hours. This occurs even as $\frac{4}{6} \approx 66.7\%$ bits are recovered. With additional erasure bits being introduced, the reconstruction time increases significantly, demonstrating the efficiency advantages of our misaligned scenarios.

Table 6: Comparison of the average number of incorrect partial solutions examined in theoretical estimates and experimental results.

		$\tau \equiv 0$	$\tau \equiv 1, 5$	$\tau \equiv 2, 3, 4$
RSA-2048	Theoretical	1,579	683	0
	Practical	1,739	640	0
	Occurrence	34.4%	35.9%	29.7%
RSA-4096	Theoretical	3,158	1,366	0
	Practical	3,484	1,278	0
	Occurrence	34.5%	36%	29.5%

6 Conclusion and Implications

In this work, we have demonstrated that the complexity of the branch-and-prune algorithm is closely tied to observed leakage patterns, and identified a new vulnerability in OpenSSL's implementation of Miller-Rabin primality test. Our proposed leakage model transforms the extracted bits into an exploitable form. Complexity analysis shows that these leakage patterns not only enable highly efficient key recovery in certain scenarios but also suggest a tolerance for additional erasure bits.

We emphasize that targeting the modular exponentiation invoked in the Miller–Rabin test offers several advantages:

- It benefits from our novel leakage patterns;
- It enables direct recovery of partial bits of p and q, bypassing complications associated with CRT-based implementations;

Moreover, repeated exponentiation calls improve the reliability of recovered information. Although not utilized in this work since our SPA attack requires only a single execution of modular exponentiation, this redundancy may serve as a valuable asset for attacking noisier environments.

Finally, this work operates under an erasure model, assuming the correctness of extracted secret bits, and has been validated on an ARM Cortex-M4 platform. Lattice-based methods are not employed due to the low complexity of the proposed attacks. Future work may extend this approach to an error model, potentially combining lattice-based techniques with the recovery of the lower 50% of bits to enhance attack efficiency.

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